

Minkyu Jun

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EDUCATION

University of Michigan — Ann Arbor

Ann Arbor, MI

Master of Science in Engineering, Computer Science (GPA: 4.00)

Expected Dec 2026

Bachelor of Science in Engineering, Computer Science (GPA: 3.98)

Dec 2025

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Scalable Systems, Operating Systems, Web Systems, Database Management Systems, Software Engineering

SKILLS

- **Languages:** Go, Python, C++, C#, Java, JavaScript, TypeScript, HTML5, CSS, SQL
- **Technologies and Frameworks:** React, React Native, Flask, PostgreSQL, MongoDB, NumPy, PyTorch, AWS
- **Developer Tools:** Git, GitHub, GitLab, Linux, Visual Studio Code, IntelliJ, Jira, Figma

WORK EXPERIENCE

Uber Freight

Chicago, IL

Software Engineer Intern

Jun 2026 — Aug 2026

- Engineered a bidirectional **gRPC** streaming service in **Go** using goroutine-based concurrency to simultaneously stream self-service budget Excel uploads to Uber's distributed storage system and an AI-assisted schema-mapping agent.
- Built an **AI-assisted schema-mapping engine** that generates context-aware prompts to map arbitrary user headers to a canonical schema, achieving **93%** mapping accuracy with multi-pass validation and per-row, per-column error diagnostics.
- Designed and constructed the budget schema and ingestion pipeline in **Google Cloud BigQuery**, including data normalization, transactional idempotent upserts, and queries joining with shipment table for budget-versus-actual cost analysis across **5000+** records.

Epic Systems

Verona, WI

Software Developer Intern

May 2025 — Aug 2025

- Developed an AI agent using OpenAI API, prompt engineering, and **Elasticsearch** vector search to automate natural language-to-JSON schema conversion, reducing scheduling rule configuration time by **75%** with **90%** accuracy.
- Integrated the AI agent into the system's front end by building a **responsive UI** with **React/TypeScript** that dynamically mapped AI-generated JSON to interface components, visualizing the AI agent's output in real-time.
- Created a **C# (.NET)** backend orchestration layer for an AI support chatbot that dynamically routes user queries to 4 specialized AI agents, providing rapid configuration guidance for users.

University of Michigan Center for Healthcare Engineering & Patient Safety (CHEPS)

Ann Arbor, MI

Software Engineering Intern

Aug 2024 — Jan 2025

- Implemented linear programming with **C++** to build a scheduling optimization tool that prioritized a hierarchy of metrics, reducing manual scheduling time for **250+** Michigan Medicine staffs by **70%**
- Migrated from IBM CPLEX solver to the in-house OptiCore library, improving maintainability across modules.

PROJECT EXPERIENCE

Multimodal AI-Generated Video Detector | Python, PyTorch, ResNet-18

- Architected a multimodal AI-generated video detector in **PyTorch**, leveraging dual **ResNet-18** encoders for audio spectrogram and video frame feature extraction and a late-fusion MLP to identify real/fake audio-video pairs.
- Achieved **97%** in both accuracy and F1 score on the 3000+ videos from the Multiple Visual Artifact Detector (MVAD) dataset, outperforming unimodal baseline model by 7%.

Open Course | React, JavaScript, Flask, Python, PostgreSQL

- Built a server-side CRUD REST API using **Flask** and **PostgreSQL** implementing query filtering and data transactions that allows users to create profiles, post group study plans, and join existing groups.
- Crafted dynamic **React/Tailwind** frontend with React Hooks and conditional rendering, enabling users to browse and filter study group posts efficiently, improving overall user engagement.

TEACHING EXPERIENCE

University of Michigan Department of Computer Science & Engineering

Ann Arbor, MI

Database Management System Teaching Assistant

Aug 2025 — Present

- Assist in teaching an upper-level computer science course with **300+** students, leading a weekly discussion session with **30+** students covering ER diagrams, **SQL**, **Java**, **MongoDB**, query optimization, sorting techniques, and crash recovery.